

## SUKHOTHAI, Thailand

WMO: 483720

Lat: 17.1069N

Lon: 99.8003E

Elev: 50

StdP: 100.73

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |             |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|-------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |             |             | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>16.8 | (c)<br>18.5 | (d)<br>12.7                   | (e)<br>9.2 | (f)<br>20.6 | (g)<br>14.9 | (h)<br>10.6 | (i)<br>22.7 | (j)<br>3.5            | (k)<br>24.2 | (l)<br>3.3 | (m)<br>25.1 | (n)<br>0.5            | (o)<br>N/A | (p)<br>0.318 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>4      | (b)<br>10.2            | (c)<br>38.9     | (d)<br>27.0 | (e)<br>37.6 | (f)<br>27.0 | (g)<br>36.3 | (h)<br>26.8 | (i)<br>28.8         | (j)<br>35.3 | (k)<br>28.4 | (l)<br>34.2 | (m)<br>28.0 | (n)<br>33.4 | (o)<br>1.4           | (p)<br>N/A |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 27.4                            | 23.5 | 31.3 | 27.1 | 23.0 | 30.8 | 26.7 | 22.5 | 30.4 | 94.4          | 35.5 | 92.1 | 34.2 | 90.3 | 33.5 | 30.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |  |  |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|--|--|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |  |  |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |  |  |
| (a) 5.3           | (b) 4.2 | (c) 3.3 | (e) 14.5                   | (f) 39.6 | (g) 2.1            | (h) 1.6 | (i) 13.0                                           | (j) 40.8 | (k) 11.8   | (l) 41.8 | (m) 10.7   | (n) 42.7 | (o) 9.2    | (p) 43.9 |  |  |
|                   |         |         | DB                         | WB       | DB                 | WB      | DB                                                 | WB       | DB         | WB       | DB         | WB       | DB         | WB       |  |  |
|                   |         |         | 13.8                       | 29.6     | 2.5                | 0.6     | 12.0                                               | 30.1     | 10.6       | 30.4     | 9.2        | 30.8     | 7.5        | 31.3     |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 28.4   | 25.2  | 27.0  | 29.0  | 31.2  | 30.8  | 29.6  | 29.1  | 28.8  | 29.0  | 28.8  | 27.5  | 25.4  |
|                                                                        | DBStd       | 2.50   | 2.01  | 1.68  | 2.24  | 2.09  | 2.28  | 1.35  | 1.37  | 1.03  | 1.14  | 1.24  | 1.64  | 2.17  |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD18.3     | 1      | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD10.0     | 6731   | 472   | 476   | 588   | 636   | 645   | 588   | 591   | 583   | 569   | 583   | 524   | 477   |
|                                                                        | CDD18.3     | 3689   | 214   | 242   | 329   | 386   | 387   | 338   | 332   | 324   | 319   | 324   | 274   | 219   |
|                                                                        | CDH23.3     | 42751  | 1797  | 2477  | 4068  | 5369  | 5161  | 4109  | 3901  | 3687  | 3697  | 3774  | 2828  | 1883  |
|                                                                        | CDH26.7     | 19459  | 627   | 1102  | 2133  | 3129  | 2821  | 1831  | 1599  | 1401  | 1492  | 1549  | 1106  | 671   |
| (14) Wind                                                              | WSAvg       | 0.7    | 0.3   | 0.4   | 0.8   | 1.1   | 1.0   | 0.9   | 1.1   | 0.8   | 0.6   | 0.4   | 0.5   | 0.4   |
| (15) Precipitation                                                     | PrecAvg     | 1209   | 10    | 12    | 30    | 63    | 189   | 153   | 141   | 185   | 243   | 149   | 28    | 9     |
|                                                                        | PrecMax     | 1754   | 54    | 71    | 160   | 161   | 382   | 264   | 259   | 263   | 365   | 275   | 112   | 70    |
|                                                                        | PrecMin     | 860    | 0     | 0     | 0     | 7     | 33    | 47    | 81    | 102   | 131   | 34    | 0     | 0     |
|                                                                        | PrecStd     | 211    | 14    | 19    | 38    | 41    | 108   | 56    | 46    | 54    | 69    | 67    | 30    | 16    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 33.1  | 35.4  | 38.6  | 40.3  | 41.4  | 37.1  | 36.0  | 34.3  | 35.1  | 35.1  | 34.3  | 34.0  |
|                                                                        |             | MCWB   | 24.5  | 24.8  | 24.3  | 27.6  | 28.5  | 27.1  | 27.1  | 27.6  | 27.9  | 27.5  | 26.8  | 25.5  |
|                                                                        | 2%          | DB     | 32.0  | 34.2  | 37.0  | 38.9  | 39.4  | 35.2  | 34.3  | 33.2  | 34.0  | 34.0  | 33.3  | 32.6  |
|                                                                        |             | MCWB   | 24.0  | 25.0  | 25.2  | 27.3  | 27.8  | 27.2  | 26.9  | 27.1  | 27.5  | 27.2  | 26.3  | 24.5  |
|                                                                        | 5%          | DB     | 31.1  | 33.2  | 35.8  | 37.7  | 37.6  | 34.0  | 33.2  | 32.5  | 33.1  | 33.1  | 32.5  | 31.6  |
|                                                                        |             | MCWB   | 23.8  | 25.0  | 25.3  | 27.3  | 27.5  | 27.2  | 26.7  | 26.9  | 27.3  | 27.0  | 25.9  | 24.4  |
|                                                                        | 10%         | DB     | 30.1  | 32.1  | 34.7  | 36.5  | 35.9  | 32.9  | 32.2  | 31.7  | 32.2  | 32.3  | 31.6  | 30.3  |
|                                                                        |             | MCWB   | 23.5  | 24.7  | 25.4  | 27.2  | 27.4  | 27.0  | 26.5  | 26.8  | 27.1  | 26.8  | 25.6  | 24.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 26.4  | 27.6  | 28.0  | 29.4  | 29.7  | 28.6  | 28.2  | 28.6  | 28.7  | 28.6  | 28.0  | 26.8  |
|                                                                        |             | MCDB   | 30.8  | 32.3  | 34.2  | 37.2  | 37.0  | 33.2  | 33.0  | 32.8  | 33.4  | 33.0  | 32.4  | 31.1  |
|                                                                        | 2%          | WB     | 25.7  | 26.8  | 27.2  | 28.7  | 29.0  | 28.0  | 27.6  | 28.0  | 28.2  | 28.0  | 27.2  | 26.1  |
|                                                                        |             | MCDB   | 30.0  | 31.5  | 33.7  | 36.3  | 36.3  | 32.7  | 32.3  | 31.8  | 32.4  | 32.0  | 31.5  | 30.4  |
|                                                                        | 5%          | WB     | 25.2  | 26.2  | 26.7  | 28.1  | 28.4  | 27.7  | 27.2  | 27.5  | 27.7  | 27.6  | 26.9  | 25.4  |
|                                                                        |             | MCDB   | 29.1  | 31.1  | 33.1  | 35.6  | 34.9  | 32.2  | 31.6  | 31.2  | 31.7  | 31.5  | 30.9  | 29.4  |
|                                                                        | 10%         | WB     | 24.5  | 25.7  | 26.2  | 27.6  | 27.9  | 27.3  | 26.9  | 27.1  | 27.4  | 27.2  | 26.4  | 24.7  |
|                                                                        |             | MCDB   | 28.1  | 30.4  | 32.3  | 34.6  | 33.8  | 31.7  | 30.9  | 30.7  | 31.2  | 31.0  | 30.0  | 28.6  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 10.0  | 11.1  | 11.1  | 10.2  | 8.7   | 6.9   | 6.3   | 6.1   | 6.6   | 6.9   | 8.3   | 9.7   |
|                                                                        | 5% DB       | MCDBR  | 11.0  | 12.2  | 12.7  | 11.4  | 11.3  | 8.3   | 7.7   | 7.2   | 7.9   | 8.2   | 9.1   | 10.7  |
|                                                                        |             | MCWBR  | 4.3   | 4.6   | 4.2   | 3.5   | 3.3   | 2.8   | 2.9   | 2.9   | 2.9   | 2.5   | 3.0   | 3.8   |
|                                                                        | 5% WB       | MCDBR  | 9.1   | 10.7  | 11.1  | 10.6  | 9.5   | 7.4   | 6.9   | 6.8   | 7.4   | 7.5   | 8.3   | 9.4   |
|                                                                        |             | MCWBR  | 4.1   | 4.6   | 4.1   | 3.8   | 3.5   | 3.0   | 2.8   | 3.0   | 3.0   | 2.8   | 3.0   | 3.5   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.476 | 0.573 | 0.678 | 0.628 | 0.500 | 0.443 | 0.432 | 0.440 | 0.479 | 0.548 | 0.469 | 0.470 |
|                                                                        | taud        |        | 2.095 | 1.863 | 1.652 | 1.808 | 2.196 | 2.380 | 2.410 | 2.373 | 2.251 | 2.007 | 2.205 | 2.135 |
|                                                                        | Ebn at Noon |        | 804   | 744   | 682   | 721   | 809   | 848   | 859   | 859   | 828   | 755   | 807   | 797   |
|                                                                        | Edh at Noon |        | 154   | 203   | 258   | 222   | 148   | 122   | 119   | 124   | 140   | 173   | 137   | 144   |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 4.66  | 5.21  | 5.45  | 6.00  | 5.55  | 5.00  | 4.68  | 4.58  | 4.68  | 4.68  | 4.77  | 4.49  |
|                                                                        | RadStd      |        | 0.24  | 0.22  | 0.33  | 0.31  | 0.44  | 0.38  | 0.32  | 0.26  | 0.26  | 0.28  | 0.26  | 0.19  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (17 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | +0.14       | N/A     | N/A     | +103    |

Nomenclature: See separate page